Inequalities — problem 7.65

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$a + b(x + y + z + u) + c(xy + xz + xt + yz + yt + zt) + d(xyz + xyt + xzt + yzt) + exyzt \geq 0$, $0 \leq x, y, z, t \leq 1$ iff $a \geq 0$; $a + b \geq 0$; $a + 2b + c \geq 0$, $a + 3b + 3c + d \geq 0$, $a + 4b + 6c + 4d + 4 \geq 0$.